

VESANTO SONKARI, Finland

WMO: 027100

Lat: 62.922N

Lon: 26.425E

Elev: 394

StdP: 14.49

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-11.3	-22.6	1.6	-17.2	-16.0	2.3	-11.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	81.2	65.4	77.4	63.5	74.0	61.7	68.2	75.9	66.2	74.3	64.3	71.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.9	71.0	63.5	89.1	69.4	61.3	82.2	67.4	32.7	75.9	31.1	74.3	29.7	71.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-20.9	85.3	8.6	4.8	-27.1	88.8	-32.2	91.6	-37.0	94.3	-43.3	97.8
			WB	-21.2	70.5	8.5	3.7	-27.3	73.2	-32.3	75.4	-37.1	77.5	-43.2	80.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	14.1	18.3	25.3	36.4	48.6	55.8	61.9	58.2	49.7	37.5	31.0	22.5		
	DBStd	18.08	13.73	13.26	9.26	6.54	8.34	7.23	6.37	5.80	5.95	6.58	7.74	11.85		
	HDD50	5223	1114	888	764	409	127	24	1	6	77	391	569	853		
	HDD65	9786	1579	1308	1229	859	510	286	141	223	460	854	1019	1318		
	CDD50	982	0	0	0	0	84	199	368	261	68	2	0	0		
	CDD65	69	0	0	0	0	2	11	44	12	0	0	0	0		
	CDH74	754	0	0	0	0	68	145	429	112	0	0	0	0		
	CDH80	137	0	0	0	0	5	22	89	21	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	25.3	1.6	1.3	1.3	1.3	1.8	2.8	3.6	2.8	2.2	2.4	2.1	2.0		
(15) Precipitation	PrecMax	32.2	3.2	2.6	3.1	2.6	4.5	4.6	7.4	5.4	4.2	4.4	4.5	3.1		
	PrecMin	20.8	0.4	0.0	0.2	0.4	0.4	0.8	1.1	0.8	0.9	0.6	0.2	0.6		
	PrecStd	3.0	0.7	0.7	0.6	0.6	1.0	0.9	1.4	1.1	1.1	1.1	0.9	0.7		
	PrecStd	3.0	0.7	0.7	0.6	0.6	1.0	0.9	1.4	1.1	1.1	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.3	41.8	48.3	64.9	80.3	83.1	86.2	82.7	68.7	55.9	49.7	42.1		
		MCWB	35.3	37.7	39.2	48.8	59.5	65.0	66.4	67.9	59.6	51.9	48.0	41.3		
	2%	DB	34.2	38.4	43.4	56.9	75.2	78.4	82.3	76.6	65.0	52.2	45.3	38.3		
		MCWB	33.4	35.9	36.9	44.8	56.8	63.2	66.8	64.1	58.0	50.0	44.1	37.2		
	5%	DB	32.8	35.6	39.6	52.2	70.9	73.9	79.0	73.0	61.9	50.2	42.7	35.9		
		MCWB	32.3	34.6	34.8	42.7	56.2	60.6	65.6	62.1	56.2	48.1	42.1	35.0		
	10%	DB	31.7	33.3	36.9	48.1	65.3	70.2	75.1	69.6	59.2	47.4	40.2	34.1		
		MCWB	31.4	32.5	33.7	41.3	53.0	59.0	64.5	61.2	55.2	45.7	39.5	33.5		
	0.4%	WB	35.9	38.4	40.7	49.7	63.0	67.4	72.3	69.7	61.1	53.3	48.0	41.8		
		MCDB	36.4	41.0	45.5	59.2	74.8	77.6	79.2	78.9	66.8	55.3	49.3	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.7	36.2	37.9	46.8	59.6	65.2	69.3	66.2	59.2	51.0	44.3	37.4		
		MCDB	34.0	38.1	42.0	55.8	70.0	74.2	77.3	74.2	63.1	52.1	45.0	38.0		
	5%	WB	32.5	34.5	36.0	43.8	57.2	62.5	67.6	64.3	57.4	48.5	42.1	35.1		
		MCDB	32.8	35.5	38.7	50.6	68.2	71.7	75.1	70.5	60.7	49.8	42.7	35.6		
	10%	WB	31.4	32.7	34.4	41.6	54.3	60.0	65.9	62.5	55.9	45.8	39.7	33.5		
		MCDB	31.8	33.4	36.2	46.9	63.9	68.2	73.7	67.2	58.6	47.3	40.1	34.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.2	16.4	17.7	20.2	19.1	18.2	17.4	14.0	10.2	6.3	8.0		
		MCDBR	6.2	10.1	17.3	26.0	29.9	24.8	23.6	22.8	19.1	13.5	9.0	7.6		
		MCWBR	6.2	8.6	12.2	15.4	13.9	11.8	10.5	11.6	12.7	10.8	8.7	7.2		
		MCDBR	6.0	9.7	14.3	21.8	25.5	23.2	20.4	19.1	16.5	12.5	9.1	7.1		
	5% WB	MCWBR	6.0	8.4	10.8	13.8	13.0	11.5	10.5	10.7	11.4	10.8	8.9	6.8		
		MCWBR	6.0	8.4	10.8	13.8	13.0	11.5	10.5	10.7	11.4	10.8	8.9	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.228	0.284	0.311	0.353	0.348	0.347	0.371	0.355	0.331	0.311	0.246	N/A		
	taud		2.013	2.127	2.199	2.332	2.452	2.481	2.430	2.464	2.519	2.400	2.104	N/A		
	Ebn at Noon		127	204	248	263	276	278	266	261	246	200	119	N/A		
	Edh at Noon		10	21	29	32	31	30	31	28	22	16	9	N/A		
(44) All-Sky Solar Radiation	RadAvg		59	237	722	1156	1523	1666	1587	1216	683	266	67	26		
	RadStd		9	56	85	137	165	110	161	136	77	48	12	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page